

Predictive Modelling Report: NBA Score Analysis

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Take-home Challenge for Sporting Risk

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1 Introduction

This document presents two approaches to predicting team point distributions in the NBA. They rely on processing approximately 15,000 historical NBA match statistics from the year 2020, provided as part of this take-home challenge.

The original dataset provides 125 meta/macro statistics for each match. To predict the distribution for a given match, both approaches process the historical statistics of the home and away teams. These past statistics are calculated by averaging data over different rolling windows.

The goal of the first approach is to select important features and provide a team point distribution. Assuming linearity between features and team points is a simple and effective starting point. A well-known model for this is LASSO regression, which serves as our first approach.

Building on the important features identified in a linear framework, the second approach aims to capture non-linearities and provide a more accurate team point distribution. Random Forests are an excellent method to achieve this.

The primary objective throughout this task is to evaluate the models, i.e., to determine whether they outperform naive predictions (which simply assume a point distribution based on basic assumptions). Thus, the goal is to measure the extent to which these models overperform naive ones. As a performance metric, we use the Ranked Probability Score (RPS), which is highly regarded in the sports industry.

To maintain flexibility, all models are built from scratch without using scikit-learn. This also reflects how I learned data science academically at Ensimag and Imperial College London.

2 Feature Selection with LASSO

2.1 Feature Pre-selection

The goal of this section is to extract important features for score distribution prediction. As mentioned, we use past statistics for each team over different rolling windows. Given the arbitrary rolling windows of 3, 5, and 10 previous matches, the total feature set to shrink would be approximately $125 \times 3 = 375$. However, one can easily and drastically reduce this amount using common sense. With a little sports intuition, one can anticipate which variables are relevant to consider when predicting team scores. For example, average team points in past matches are likely more relevant than the number of isolation rim attempts.

The feature pre-selection process can be found in the notebook linked with this document.

2.2 Justification and Presentation of LASSO Penalization

LASSO has two main advantages for predictive modelling in sports:

- It performs feature selection, providing interpretability and reducing correlations.
- Even though it adds bias to the point estimator, it lowers its variance. This means that if one uses two slightly different training sets to build two models, the resulting models will remain close to each other. In other words, the model is less sensitive to the intrinsic, highly noisy nature of sports statistics. As long as the bias is controlled, this is a desirable property.

Given the random variable Y representing team points, and the matrix $\mathbf{X}$ of past team statistics, the linear regression problem can be written as follows:

$$Y = \mathbf{X}^T \boldsymbol{\beta} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, I\sigma^2)$$

where $\mathcal{N}(0, I\sigma^2)$ is a multivariate normal distribution with mean 0 and covariance matrix $I\sigma^2$. The fact that the covariance matrix is a multiple of the identity matrix suggests no correlation between the noise of different matches, which is a reasonable assumption. Even if there were noise correlation, since we use statistics based on past matches, the model intrinsically accounts for noise correlation between matches of the same teams, minimizing the issue.

Thus, under this model, the assumed team point distribution for a match with statistics x is $\mathcal{N}(x\beta, \sigma^2)$, and its estimate will be $\mathcal{N}(x\hat{\beta}, \hat{\sigma}^2)$, where $\hat{\sigma}^2$ is the Mean Squared Error (MSE) of the residuals.

The fact that the distribution of team points is continuous rather than discrete is a theoretical disadvantage, but we maintain this assumption for now.

In LASSO regression, the aim is to minimize the following loss function:

$$L_{\text{LASSO}}(\beta) = \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda\|\beta\|_1$$

where

$$\|\beta\|_1 = \sum_{i=1}^p |\beta_i| \quad (\text{Hence the feature selection}).$$

The gradient of this loss is:

$$\nabla_{\beta} L_{\text{LASSO}}(\beta) = 2(\mathbf{X}^T \mathbf{X}\beta - \mathbf{X}^T \mathbf{y}) + \lambda \text{sign}(\beta) \quad (1)$$

This equation has no analytical solution (it cannot simply be set to zero), so the gradient descent algorithm is used to perform the minimization:

$$\beta_{t+1} = \beta_t - \eta_t \nabla_{\beta} L_{\text{LASSO}}(\beta_t)$$

where η_t is the step size (learning rate) of the descent.

To determine the optimal λ for the model, we perform K-fold cross-validation (respecting time order to avoid data leakage) using the Ranked Probability Score (*RPS*) as the loss function.

For a single basketball match with K possible ordered outcomes, the *RPS* is defined as:

$$RPS = \sum_{k=1}^K (P_k - O_k)^2$$

This can be expanded using the individual probability mass functions:

$$RPS = \sum_{k=1}^K \left(\sum_{j=1}^k p_j - \sum_{j=1}^k o_j \right)^2$$

where K is the total number of possible outcomes (here, the possible points scored, assumed to range from 0 to 200); p_j is the predicted probability that the exact outcome is j ; o_j is the actual observed outcome (equal to 1 if the actual score is j , and 0 otherwise); and P_k and O_k are the cumulative probabilities of the forecast and the observation, respectively, at threshold k . (Thus, $\exists k^* \in \{1, \dots, K\}$ such that $\forall k, O_k = \mathbb{1}_{\{k \geq k^*\}}$, meaning O is a Heaviside step function).

When evaluating the model (first during cross-validation, then on the test set), we average the *RPS* over the samples (matches):

$$\overline{RPS} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K (P_{i,k} - O_{i,k})^2$$

where N is the number of predicted team point distributions (i.e., the number of predicted matches).

For cross-validation, we use the mean of the averaged *RPS* across the folds.

The *RPS* offers two main advantages:

- It evaluates the entire probability distribution rather than a single point estimate, rewarding models that accurately quantify their uncertainty and are well-calibrated.
- It respects the ordinal nature of the outcomes by measuring the "distance" of an error. Because it uses cumulative probabilities, predicting 109 points when the true score is 110 is penalized much less than predicting 80 points, which is highly relevant for sports forecasting.

To evaluate the true contribution of the algorithms, analyzing the raw *RPS* score alone is insufficient, as its absolute value is difficult to interpret in isolation. The "Skill Score" quantifies the percentage improvement of a predictive model compared to a reference forecast. Here, the naive forecast predicts the point distribution as a discretized normal distribution, with a mean equal to the team's average points over the past 10 matches, and a variance equal to the mean squared error of past residuals.

$$RPSS = 1 - \frac{\overline{RPS}_{\text{model}}}{\overline{RPS}_{\text{naive}}}$$

2.3 Results

The 7-fold cross-validation identified the optimal penalty at $\lambda = 6773$.

At this λ , the selected important features are:

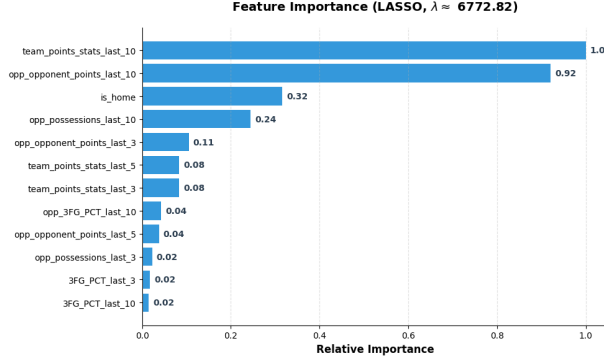


Figure 1: Feature importance plot

Unsurprisingly, the most important feature is the average team points over past matches, which reflects the team’s offensive strength. The second most important feature evaluates the weakness of the opponent’s defense. The third is whether the team in question is playing at home or not, which is a well-known critical factor.

Finally, when computing RPS_{model} and RPS_{naive} on the test set, we obtain the following $RPSS$:

$$RPSS = 6.61\%$$

Such a positive $RPSS$ means that the model provides better distributions (in terms of RPS comparison) than the naive one. A score above 5% suggests solid performance.

3 Random Forests

3.1 Justification

A limitation of LASSO is the linearity assumption, which can be restrictive. Furthermore, the primary aim of such regressions is to predict a single point estimate, not a distribution. To address this, we can implement decision trees to produce empirical distributions of team points, trained by minimizing the variance of each distribution. Because decision trees tend to overfit, we mitigate this by implementing Random Forests. This method trains multiple trees by randomly sampling both the training set (bootstrapping) and the features on which to split the data (feature bagging).

As with LASSO, we perform cross-validation using RPS to determine the optimal value for a key hyperparameter: the minimum standard deviation σ_{min} of team points within the leaves of the trees. This means we stop splitting a node when this standard deviation threshold is reached.

3.2 Results

The cross-validation provides the following plot:

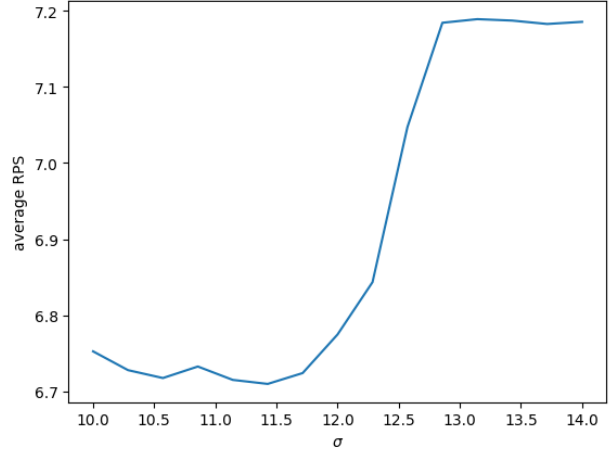


Figure 2: Forest loss curve

The cross-validation indicates an optimal σ of around 11.5.

To reduce the variance of the aggregated predictions, the final forest was retrained with $B = 200$ trees (the number of trees used during cross-validation was kept lower, at $B = 15$, for computational reasons, since the optimal σ is not expected to be materially affected by B). Comparing this final model against the naive model on the test set, we obtain:

$$RPSS = 6.2\%$$

4 Conclusion and Limitations

Comparing the two models, the Random Forest is not competitive with LASSO: even after increasing the forest size to $B = 200$ trees, its skill score ($RPSS = 6.2\%$) remains below that of LASSO ($RPSS = 6.61\%$). Part of this gap is explained by a design choice: the forest is trained on the reduced set of features already selected by LASSO, rather than on the full raw feature set. As the remaining features were mostly finer-grained box-score statistics rather than the kind of richer metadata (e.g., individual player form or injury reports) that would give a non-linear model like Random Forests real additional structure to exploit beyond what LASSO already captures linearly.

More generally, both models can be considered good when compared to the naive baseline used throughout this report. However, this naive baseline

is a simple reference point, and it says little about whether either model would generate genuine alpha in practice. A more demanding and industry-relevant benchmark would be to compare model predictions against bookmaker odds, which already price in a large amount of public information and are close to an efficient market; consistently beating them, even by a small margin, is a stronger indication of real predictive skill. Additionally, the RPS here is computed over a support of 201 possible outcomes (0 to 200 points), which is considerably finer-grained than the 3-outcome (win/draw/loss) setting in which RPS is traditionally used in sports forecasting, so common rules of thumb about what constitutes a "good" skill score may not directly transfer to this setting.

In conclusion, the LASSO model proves to be an effective and comparatively lightweight approach for both predicting distributions and identifying feature importance, and remains the stronger of the two models evaluated here.